

Varun Pratap Singh

Backend Developer | Node.js, Express.js, MongoDB, Redis

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PROFESSIONAL SUMMARY

Backend Developer with 1.5+ years of experience building production Node.js, Express.js and MongoDB applications with complex business workflows. Focused on REST API design, Redis and BullMQ background processing, Socket.IO real-time systems, JWT/RBAC security, Razorpay payment flows and LLM integrations.

TECHNICAL SKILLS

Backend: Node.js, Express.js, JavaScript, TypeScript, REST APIs, JWT Authentication, RBAC, API Security

Databases: MongoDB, Mongoose, MySQL, Aggregation, Indexing, Query Optimization

Real-Time & Processing: Redis, Socket.IO (WebSockets), BullMQ, Background Jobs, Caching

Cloud & Deployment: AWS EC2, AWS S3, Linux, Nginx, PM2, GitHub Actions (CI/CD), Git

Tools & APIs: Swagger/OpenAPI, Jest, Supertest (API testing), Razorpay, Firebase, OpenAI API, Agora

PROFESSIONAL EXPERIENCE

Volvrit Inc. | Full Stack Developer | Delhi, India

May 2025 – Present

- Develop and maintain Node.js/Express.js REST APIs across 20+ client projects, secured with JWT access/refresh tokens, RBAC, OTP verification and Redis-backed rate limiting.
- Cafe-Cold (cold-chain logistics):** designed a modular REST API across 18 business areas and 28 MongoDB collections with a 6-stage shipment lifecycle; streamed live GPS and temperature data over role-specific Socket.IO channels and moved email, SMS and push notifications to BullMQ workers.
- Cafe-Cold payments:** built Razorpay payments with a wallet and ledger, transporter payouts, credit requests, withdrawals and PDF invoices.
- Calling QR (smart identity QR/NFC):** built the scan, verify and activate flow with OTP, PIN lockout and privacy-first masked calling across 45+ data models and 50+ route modules; Redis enforces alert cooldowns, daily limits and a JWT token blacklist.
- Quiz Game (main backend contributor, May 2026 – Jul 2026):** wrote the Socket.IO/Redis multiplayer engine for 5 game modes with room-based matchmaking and server-side timers, plus 80+ REST APIs with Razorpay/RazorpayX payouts.
- GFG Shooting Academy:** rebuilt the talent-scoring engine for 8 fitness and shooting tests and wrote a re-runnable data migration with dry-run mode and invalid-data detection.
- Deploy to AWS EC2 with Nginx, PM2 and GitHub Actions CI/CD, integrate LLM APIs driven by system prompts, and debug production issues across API, database and queue layers.

ToXSL Technologies | Software Developer

Jul 2024 – Oct 2024

- Developed and debugged backend modules for client web applications in Java and .NET Core.

Oasis Infobyte | Web Development Intern (Virtual)

Oct 2023 – Jan 2024

SELECTED PROJECTS

Website Builder: Multi-Tenant SaaS Platform | Node.js, Express.js, MongoDB, Redis, OpenAI API | Personal Project

- Built a multi-tenant backend with tenant data isolation, RBAC and plan-based feature gating through Razorpay subscriptions with webhook verification.
- Secured sign-in with email OTP, rotating refresh tokens and a Redis token blacklist; covered the APIs with Jest and Supertest tests.
- Published sites on subdomains, paths or custom domains through Nginx with automated Let's Encrypt SSL; added OpenAI-generated content with usage quotas and images on AWS S3.

EDUCATION

B.Tech, Computer Science & Engineering, Uttarakhand Technical University

2020 – 2024 | CGPA: 7.3/10

Class XII (CBSE): 83%, 2019 | Class X (CBSE): CGPA 9.6/10, 2017 | Saraswati Vidya Mandir, Etah